

ANSH KUMAR SINGH

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Education

Indian Institute of Information Technology, Jabalpur

Bachelor of Technology in Computer Science and Engineering

Aug. 2024 – Present

Madhya Pradesh, India

Relevant Coursework

- Data Structures & Algorithms
- Database Management Systems
- Operating Systems
- Object-Oriented Programming
- Computer Architecture
- Computer Networks

Projects

Aegis | *TypeScript, Babel AST, Groq AI, MongoDB*

3rd Overall – HackByte 4.0 (MLH) | Apr. 2026

- Engineered a zero-trust CLI scanner that parses **Abstract Syntax Trees (ASTs)** via Babel to detect phantom dependencies and supply-chain threats with **sub-500 ms** execution time.
- Architected **6 parallel heuristic scanners** to detect dangerous API calls (`eval`, `spawn`), unauthorized file access, and network exfiltration attempts in **under 1 second**.
- Leveraged **Groq AI** for typosquatting classification through a **3-tier risk pipeline** (Block / Flag / Allow). **Published** to **npm registry** with under 2s total audit overhead.

Clustify | *JavaScript, OAuth 2.0, WebExtensions API, Gmail API*

1st Place – Institute Hackathon | Oct. 2024

- Developed a cross-browser extension (Chrome & Firefox) that batch-processes **3,000+ emails** via the **Gmail API**, reducing manual inbox management time to near zero.
- Designed an **AI Auto-Labeler** for rule-based email categorization and configured **Google OAuth 2.0** for encrypted, scoped access to user data.
- Shipped a **preview-before-delete interface** with non-blocking async API calls; published to **Chrome Web Store** and **Mozilla Add-ons Store**.

Technical Skills

Languages: C++, TypeScript, JavaScript, SQL, HTML5, CSS3, Python, Java, C

Frameworks & Libraries: Node.js, Express.js, Next.js, React, Tailwind CSS

Tools: Docker, Git/GitHub, Postman, PostgreSQL, MongoDB, MySQL, Groq AI

Platforms: AWS, Google Cloud Platform, IBM Cloud, Linux (Ubuntu)

Concepts: REST APIs, AST Parsing, OAuth 2.0, CI/CD, Security Auditing, Supply-Chain Analysis

Honors & Awards

HackByte 4.0 – Major League Hacking (MLH)

Apr. 2026

3rd Place Overall – 120+ Teams, 500+ Participants

IIIT Jabalpur

- Placed **3rd out of 120+ teams** nationwide; developed **Aegis**, a zero-trust npm supply-chain scanner using AST parsing and AI-driven risk classification.

“Can You Hack It” – Institute-Level Hackathon

Oct. 2024

1st Place – 400+ Participants

IIIT Jabalpur

- Won 1st place among **400+ participants** for **Clustify**, a cross-browser Gmail automation extension with AI-powered email categorization.

Newbie 1.0 – Competitive Programming

Nov. 2024

12th Rank – 500+ Participants

IIIT Jabalpur

- Finished **12th out of 500+** first-year students in institute-level competitive programming contest.

Leadership / Extracurricular

Basketball Club – IIIT Jabalpur

Aug. 2024 – Present

Senior Member & Team Lead

Jabalpur, India

- Led a team of 15+ members by planning weekly practice schedules, inter-institute match tactics, and player skill development programs.
- Directed selection trials for **100+ candidates**, managed equipment inventory, and coordinated with institute administration for facility access.